

GETRA Community Handoff

Dokumentasi final fitur Community untuk tim produk, frontend, backend, QA, dan maintainer.

Tanggal handoff: 23 Agustus 2026

Repository: `D:\Getra_Production`

Output PDF: `docs/handoff_community.pdf`

1. Ringkasan Eksekutif

GETRA Community adalah fitur sosial berbasis lokasi untuk pengguna GETRA. Fitur ini menggabungkan feed diskusi bergaya thread, temuan komuter, cultural map, permintaan kebutuhan lokal, sinyal demand teragregasi, respons UMKM, notifikasi, moderasi, reputasi, friendship, foto profil canonical, dan upload foto post yang aman.

Community dibangun dengan pola full stack yang sudah ada di GETRA:

- Frontend Next.js App Router di workspace `frontend`.
- Backend Next.js API route di workspace `backend`.
- Domain layer di `backend/src/features/community`.
- Supabase sebagai database, storage, RLS, RPC, dan realtime event source.
- Auth menggunakan authenticated GETRA/Supabase user.
- Semua response API memakai envelope GETRA standar melalui helper `createSuccessResponse` dan `createListResponse`.

Status handoff:

- Implementasi Community utama sudah tersedia dari sisi file, route, service, repository, UI, storage, dan migration.
- Backend unit test terakhir yang dijalankan lulus: 75 test file passed, 511 test passed, 1 skipped.
- Gate global project belum sepenuhnya hijau karena error typecheck/build/lint di modul lain seperti UMKM workspace, merchant submission, dan UMKM advertising. Detail ada di bagian "Known Blockers".

2. Tujuan Produk

Community dibuat untuk:

- Memberi ruang berbagi informasi antar komuter.
- Mengumpulkan temuan lokal seperti makanan, landmark, pusat kerajinan, sejarah lokal, dan aktivitas komunitas.
- Membuat peta budaya yang dapat dilihat berdasarkan bounding box peta.
- Memungkinkan komuter mengirim permintaan kebutuhan sekitar lokasi.
- Mengubah permintaan yang berulang menjadi demand signal agregat untuk pelaku UMKM.
- Memungkinkan UMKM merespons sinyal permintaan.
- Membangun trust melalui reaksi, reputasi, laporan, moderasi, notifikasi, dan pertemanan.

3. Struktur File Utama

Backend API Route

Semua endpoint Community berada di:

```
backend/app/api/community/  
backend/app/api/admin/community/
```

Struktur route:

```

backend/app/api/community/
  cultural-map/route.ts
  feed/route.ts
  posts/route.ts
  posts/[postId]/route.ts
  posts/[postId]/comments/route.ts
  posts/[postId]/reactions/[reactionType]/route.ts
  requests/route.ts
  requests/[requestId]/route.ts
  requests/signals/route.ts
  requests/signals/[signalId]/route.ts
  requests/signals/[signalId]/responses/route.ts
  notifications/route.ts
  notifications/[notificationId]/read/route.ts
  notifications/read-all/route.ts
  reports/route.ts
  reputation/[userId]/route.ts
  users/[userId]/route.ts
  friends/route.ts
  friends/requests/route.ts
  friends/[friendshipId]/route.ts

backend/app/api/admin/community/
  analytics/route.ts
  reports/route.ts
  reports/[reportId]/route.ts

```

Backend Domain Layer

```

backend/src/features/community/
  constants/community.constants.ts
  index.ts
  mappers/community.mapper.ts
  repositories/community.repository.ts
  schemas/community.schema.ts
  services/community.service.ts
  services/community-media.service.ts
  types/community.types.ts

```

Tanggung jawab:

- `constants` : limit, enum, bucket, ukuran file, rule kategori, dan rule request.
- `schemas` : validasi input Zod untuk route dan service.
- `types` : contract domain TypeScript.
- `mappers` : mapping row database/RPC ke DTO frontend.
- `repository` : adapter Supabase RPC dan storage signed URL.
- `service` : validasi, orchestration, error mapping.
- `community-media.service` : validasi dan normalisasi foto post dengan Sharp.

Frontend Route

```

frontend/app/community/
  page.tsx
  loading.tsx
  error.tsx
  [id]/page.tsx
  requests/[requestId]/page.tsx
  requests/signals/[signalId]/page.tsx

```

```
users/[userId]/page.tsx
friends/page.tsx
```

Frontend Feature Layer

```
frontend/src/features/community/
  api/community.api.ts
  constants/community.constants.ts
  hooks/
    use-community-feed.ts
    use-community-post-detail.ts
    use-commuter-requests.ts
    use-demand-signals.ts
    use-demand-signal-detail.ts
    use-community-user-profile.ts
    use-community-friends.ts
  services/community-realtime.service.ts
  types/community.types.ts
  utils/community-format.ts
  components/
    community-shell.tsx
    community-navigation.tsx
    community-page.tsx
    community-empty-state.tsx
    common/community-avatar.tsx
  post/
  comments/
  feed/
  media/
  location/
  map/
  request/
  demand/
  response/
  notifications/
  moderation/
  profile/
  friends/
  community.module.css
```

4. Modul dan Flow Fitur

4.1 Beranda Community

Halaman utama berada di `frontend/app/community/page.tsx` dan memakai `CommunityPage`.

Fungsi:

- Menampilkan composer post.
- Menampilkan feed post umum dan finding.
- Mendukung foto profil canonical dari `profiles.avatar_url`.
- Mendukung upload foto opsional untuk post.
- Mendukung lokasi opsional dan privacy `APPROXIMATE` atau `EXACT`.
- Mendukung filter "Beranda" dan "Temuan Komuter".
- Mendukung navigasi ke detail post/thread.

Komponen utama:

- `CommunityShell`

- `CommunityNavigation`
- `PostComposer`
- `CommunityFeed`
- `PostCard`
- `ReactionBar`
- `CommunityAvatar`
- `CommunityNotificationsMenu`

4.2 Temuan Komuter

Temuan adalah post bertipe `FINDING`. Aturan penting:

- `type` harus `FINDING`.
- `category` wajib.
- `location` wajib.
- Kategori yang valid:
 - `LEGENDARY_EATERY`
 - `LOCAL_FOOD`
 - `CRAFT_CENTER`
 - `LANDMARK`
 - `LOCAL_HISTORY`
 - `COMMUNITY_ACTIVITY`

Temuan muncul di feed dan juga dapat muncul di Cultural Map jika memiliki lokasi.

4.3 Cultural Map

Cultural Map berada di tab "Cultural Map" dan komponen `CulturalMap`.

Fungsi:

- Mengambil finding berdasarkan bounding box peta.
- Mendukung filter kategori.
- Menampilkan marker/list temuan.
- Menampilkan avatar author, nama author, kategori, jumlah konfirmasi, jumlah reply, dan waktu.

API memakai bounding box:

```
GET /api/community/cultural-map?west=...&south=...&east=...&north=...&categories=...&limit=...
```

4.4 Permintaan Komuter

Permintaan komuter berada di tab "Permintaan".

Fungsi:

- Membuat demand/request berbasis lokasi.
- Menampilkan daftar request terbaru.
- Mendukung kategori, budget maksimum, radius, masa berlaku, dan lokasi.
- Mendukung detail request.

Kategori:

- `FOOD`
- `DRINK`
- `DAILY_NEEDS`

- SERVICE
- OTHER_LOCAL_NEED

Rule:

- Title maksimal 120 karakter.
- Description maksimal 500 karakter.
- Budget maksimal Rp10.000.000.
- Radius minimal 100 meter dan maksimal 5.000 meter.
- Opsi umum radius: 500, 1000, 2000, 3000 meter.
- Expiry: 1, 3, atau 7 hari.

4.5 Demand Signal

Demand signal adalah agregasi permintaan komuter yang mirip secara kategori, lokasi, budget, dan waktu.

Konstanta penting:

- Window agregasi: 7 hari.
- Radius cluster: 1000 meter.
- Minimal request untuk signal: 3.
- Bucket budget: Rp25.000.

UI:

- Tab Permintaan memiliki segment "Permintaan Terbaru" dan "Sinyal Community".
- Detail signal berada di `frontend/app/community/requests/signals/[signalId]/page.tsx`.

4.6 Respons UMKM

UMKM dapat merespons demand signal dengan status:

- AVAILABLE
- WILL_TRY
- PREPARING
- UNAVAILABLE

Payload respons:

```
{
  "merchant_id": "uuid",
  "status": "AVAILABLE",
  "message": "Opsional, maksimal 500 karakter"
}
```

Respons disimpan/insert per signal dan merchant.

4.7 Thread, Reply, dan Reaksi

Community mendukung:

- Detail post.
- Komentar root.
- Nested reply terbatas sampai depth 2.
- Reaction:
 - HELPFUL
 - INTERESTING

- `CONFIRMED`

Reaction `CONFIRMED` berperan dalam reputasi dan trust signal.

4.8 Notifikasi

Notifikasi mendukung tipe:

- `POST_REPLY`
- `COMMENT_REPLY`
- `POST_CONFIRMED`
- `UMKM_RESPONSE`
- `FRIEND_REQUEST`
- `FRIEND_ACCEPTED`

Notifikasi ditampilkan melalui `CommunityNotificationsMenu`.

Data actor memakai:

- `actorUserId`
- `actorDisplayName`
- `actorAvatarUrl`

`actorAvatarUrl` berasal dari canonical profile photo `profiles.avatar_url`.

4.9 Report dan Moderasi

User dapat membuat laporan untuk:

- `POST`
- `COMMENT`
- `UMKM_RESPONSE`

Reason:

- `SPAM`
- `INCORRECT_INFORMATION`
- `INVALID_PRICE`
- `INAPPROPRIATE_CONTENT`
- `WRONG_LOCATION`
- `DUPLICATE`

Admin dapat melihat report dan melakukan action:

- `HIDE`
- `REMOVE`
- `RESTORE`
- `DISMISS`

4.10 Reputasi

Reputasi user tersedia melalui endpoint:

```
GET /api/community/reputation/[userId]
```

Data:

- `confirmedContributions`

- `helpfulReceived`
- `findingsCount`
- `reputationLabel`

4.11 Friendship

Friendship mendukung:

- Lihat daftar teman.
- Lihat incoming request.
- Lihat outgoing request.
- Kirim friend request.
- Accept, decline, cancel, dan unfriend.
- State relation pada profil user:
 - `SELF`
 - `NONE`
 - `PENDING_OUTGOING`
 - `PENDING_INCOMING`
 - `FRIENDS`

Tab "Teman" sudah aktif di header Community.

4.12 Foto Profil Canonical

Community tidak menyimpan avatar sendiri. Sumber canonical adalah:

```
profiles.avatar_url
```

Upload avatar profile berada di:

```
backend/app/api/profile/avatar/route.ts
```

Bucket:

```
avatars
```

Community memakai avatar canonical pada:

- Composer post.
- Feed/post card.
- Detail post.
- Comments/replies.
- User profile.
- Friends list.
- Incoming/outgoing friend request.
- Notifications.
- Request card.
- Cultural map.

Fallback avatar memakai initials dari display name jika URL kosong atau image gagal dimuat.

5. Daftar API

Base path backend sesuai konfigurasi `NEXT_PUBLIC_API_URL` .

5.1 Public Authenticated Community API

Method	Endpoint	Fungsi	Auth
GET	/api/community/feed	List feed post	Authenticated
POST	/api/community/posts	Buat post/finding, JSON atau multipart photo	Authenticated
GET	/api/community/posts/[postId]	Detail post	Authenticated
GET	/api/community/posts/[postId]/comments	List comment/reply	Authenticated
POST	/api/community/posts/[postId]/comments	Buat comment/reply	Authenticated
PUT	/api/community/posts/[postId]/reactions/[reactionType]	Tambah reaction	Authenticated
DELETE	/api/community/posts/[postId]/reactions/[reactionType]	Hapus reaction	Authenticated
GET	/api/community/cultural-map	List finding dalam bounding box	Authenticated
GET	/api/community/requests	List commuter request	Authenticated
POST	/api/community/requests	Buat commuter request	Authenticated
GET	/api/community/requests/[requestId]	Detail commuter request	Authenticated
GET	/api/community/requests/signals	List demand signal	Authenticated
GET	/api/community/requests/signals/[signalId]	Detail demand signal	Authenticated
GET	/api/community/requests/signals/[signalId]/responses	List respons UMKM dan merchant milik user	Authenticated
POST	/api/community/requests/signals/[signalId]/responses	Upsert respons UMKM	Authenticated
GET	/api/community/notifications	List notifikasi	Authenticated
PATCH	/api/community/notifications/[notificationId]/read	Tandai satu notifikasi dibaca	Authenticated
PATCH	/api/community/notifications/read-all	Tandai semua notifikasi dibaca	Authenticated
POST	/api/community/reports	Buat report	Authenticated
GET	/api/community/reputation/[userId]	Ambil reputasi user	Authenticated
GET	/api/community/users/[userId]	Profil Community dan relationship state	Authenticated
GET	/api/community/friends	List friends/incoming/outgoing	Authenticated
POST	/api/community/friends/requests	Kirim friend request	Authenticated
PATCH	/api/community/friends/[friendshipId]	Action friendship	Authenticated

5.2 Admin Community API

Method	Endpoint	Fungsi	Auth
GET	/api/admin/community/analytics	Statistik operasional Community	Admin
GET	/api/admin/community/reports	List report untuk moderasi	Admin
PATCH	/api/admin/community/reports/[reportId]	Moderasi target report	Admin

6. Kontrak Query dan Payload

Feed

```
GET /api/community/feed?page=1&limit=20&type=GENERAL&category=LOCAL_FOOD
```

Query:

- `page` : integer, default 1.
- `limit` : integer, 1 sampai 50, default 20.
- `type` : `GENERAL` atau `FINDING` , opsional.
- `category` : finding category, opsional.

Create Post

JSON:

```
{
  "type": "GENERAL",
  "content": "Apa yang kamu temukan di sekitar kamu?",
  "location": {
    "longitude": 106.8,
    "latitude": -6.2,
    "visibility": "APPROXIMATE",
    "accuracy_m": 50
  }
}
```

Finding:

```
{
  "type": "FINDING",
  "content": "Warung legendaris dekat stasiun.",
  "category": "LEGENDARY_EATERY",
  "location": {
    "longitude": 106.8,
    "latitude": -6.2,
    "visibility": "APPROXIMATE"
  }
}
```

Multipart:

- `payload` : JSON string seperti payload di atas.
- `photo` : file `jpeg` , `png` , atau `webp` .

Comments

```
{
  "content": "Setuju, tempat ini ramai pagi.",
  "parent_comment_id": "uuid opsional"
}
```

Commuter Request

```
{
  "title": "Butuh sarapan dekat halte",
  "description": "Ada rekomendasi bubur/roti pagi?",
  "category": "FOOD",
  "max_budget": 25000,
  "location": {
    "longitude": 106.8,
    "latitude": -6.2,
    "visibility": "APPROXIMATE"
  },
  "radius_meters": 1000,
  "expires_in_days": 7
}
```

Demand Signal Response

```
{
  "merchant_id": "uuid",
  "status": "AVAILABLE",
  "message": "Kami tersedia jam 06.00-10.00."
}
```

Report

```
{
  "target_type": "POST",
  "target_id": "uuid",
  "reason": "WRONG_LOCATION",
  "details": "Lokasinya seharusnya di sisi timur stasiun."
}
```

Friendship Action

```
{
  "action": "ACCEPT"
}
```

Action valid:

- ACCEPT
- DECLINE
- CANCEL
- UNFRIEND

7. Response Envelope

Single response memakai pola:

```
{
  "success": true,
  "data": {}
}
```

List response memakai pola:

```
{
  "success": true,
  "data": [],
  "meta": {
    "page": 1,
    "limit": 20,
    "total": 100
  }
}
```

Notification response khusus berisi:

```
{
  "success": true,
  "data": {
    "items": [],
    "page": 1,
    "limit": 20,
    "total": 0,
    "unreadCount": 0
  }
}
```

Error memakai envelope standar GETRA:

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "..."
  }
}
```

8. Database dan Migration

Migration Community utama:

```
backend/supabase/migrations/20260823010000_create_community_posts.sql
backend/supabase/migrations/20260823113000_add_community_post_location.sql
backend/supabase/migrations/20260823143000_add_community_media.sql
backend/supabase/migrations/20260823170000_add_public_profile_fields.sql
backend/supabase/migrations/20260823171000_add_community_discussion_reactions.sql
backend/supabase/migrations/20260823173000_add_community_findings_cultural_map.sql
backend/supabase/migrations/20260823190000_add_commuter_requests.sql
backend/supabase/migrations/20260823193000_create_profile_avatars_bucket.sql
backend/supabase/migrations/20260823200000_add_community_demand_signals.sql
backend/supabase/migrations/20260823210000_add_community_operational_layer.sql
backend/supabase/migrations/20260823230000_add_community_friendship.sql
backend/supabase/migrations/20260824020000_community_final_avatar_projection.sql
```

Tabel Community

Tabel	Fungsi
community_posts	Post umum dan finding
community_media	Metadata foto post
community_comments	Comment dan nested reply
community_reactions	Reaction user terhadap post
commuter_requests	Permintaan kebutuhan berbasis lokasi
community_demand_signals	Cluster/agregasi demand
community_demand_signal_members	Relasi request ke signal
community_umkm_responses	Respons merchant terhadap signal
community_notifications	Notifikasi user
community_reports	Laporan konten
community_realtime_events	Event insert untuk realtime client
community_friendships	Relasi pertemanan

Storage Bucket

Bucket	Fungsi	Akses
community-media	Foto post Community	Signed URL 10 menit
avatars	Foto profil canonical	Public read, owner write via profile flow

RPC Utama

Repository memakai Supabase RPC berikut:

RPC	Fungsi
<code>create_community_post_v4</code>	Buat post/finding dan metadata media
<code>list_community_feed_v4</code>	List feed
<code>get_community_post_detail_v2</code>	Detail post
<code>list_community_comments_v1</code>	List comment/reply
<code>create_community_comment_v1</code>	Buat comment/reply
<code>add_community_reaction_v1</code>	Tambah reaction
<code>remove_community_reaction_v1</code>	Hapus reaction
<code>list_community_cultural_map_v1</code>	List finding untuk peta
<code>create_commuter_request_v1</code>	Buat commuter request
<code>list_commuter_requests_v1</code>	List commuter request
<code>get_commuter_request_detail_v1</code>	Detail commuter request
<code>list_community_demand_signals_v1</code>	List demand signal
<code>get_community_demand_signal_detail_v1</code>	Detail demand signal
<code>list_community_demand_signal_responses_v1</code>	List respons UMKM
<code>list_community_response_merchants_v1</code>	Merchant milik user yang boleh merespons
<code>upsert_community_demand_signal_response_v1</code>	Upsert respons UMKM
<code>list_community_notifications_v1</code>	List notifikasi dengan actor avatar
<code>count_community_unread_notifications_v1</code>	Hitung unread
<code>mark_community_notification_read_v1</code>	Mark single read
<code>mark_all_community_notifications_read_v1</code>	Mark all read
<code>create_community_report_v1</code>	Buat report
<code>list_admin_community_reports_v1</code>	Admin list report
<code>moderate_community_target_v1</code>	Admin moderation action
<code>get_community_reputation_v1</code>	Ambil reputasi
<code>get_community_analytics_v1</code>	Admin analytics
<code>get_community_friendship_profile_v1</code>	Profil Community plus relation state
<code>create_community_friend_request_v1</code>	Buat friend request
<code>act_on_community_friendship_v1</code>	Accept/decline/cancel/unfriend
<code>list_community_friendships_v1</code>	List friends/incoming/outgoing

9. Privacy dan Location Safety

Community mendukung dua mode lokasi:

- `EXACT` : koordinat asli.
- `APPROXIMATE` : koordinat dibulatkan atau digrid pada layer database/RPC.

Konstanta frontend/backend:

```
COMMUNITY_LOCATION_APPROXIMATE_GRID_DEGREES = 0.001
COMMUNITY_LOCATION_MAX_ACCURACY_M = 100000
```

Catatan handoff:

- Untuk tampilan publik, gunakan lokasi dari DTO yang sudah diproses oleh RPC, bukan raw input.
- Jangan expose `location_accuracy_m` kecuali untuk kebutuhan internal/debug.
- Demand signal memakai center agregat, bukan lokasi personal satu user.

10. Media Upload

Foto post Community:

- Input format: JPEG, PNG, WEBP.
- Animated image ditolak.
- Maksimal input: 10 MB.
- Maksimal pixel input: 25.000.000.
- Output dinormalisasi menjadi WEBP.
- Maksimal dimensi output: 2048 px.
- Output disimpan di bucket `community-media`.
- Path:

```
{userId}/{postId}/{mediaId}.webp
```

Jika pembuatan post gagal setelah upload, service mencoba menghapus file yang sudah terupload.

Foto profil:

- Tidak memakai `community-media`.
- Source canonical tetap `profiles.avatar_url`.
- Bucket profile avatar adalah `avatars`.

11. Realtime

Frontend service:

```
frontend/src/features/community/services/community-realtime.service.ts
```

Source event:

```
public.community_realtime_events
```

Filter yang didukung:

- `post_id=eq.{postId}`

- `signal_id=eq.{signalId}`
- `recipient_user_id=eq.{recipientUserId}`

Channel:

- `community-post-{postId}`
- `community-signal-{signalId}`
- `community-notifications-{recipientUserId}`

Service mengambil access token browser dan menjalankan `supabase.realtime.setAuth(token)` sebelum subscribe.

12. UI dan Tampilan

Navigasi Community

Tab aktif:

- Beranda
- Temuan Komuter
- Cultural Map
- Permintaan
- Teman

Komponen:

```
frontend/src/features/community/components/community-navigation.tsx
```

Beranda

Elemen tampilan:

- Composer dengan avatar user.
- Textarea konten.
- Emoji picker.
- Tambah lokasi.
- Kamera/Galeri.
- Counter 0/500.
- Tombol posting.
- Feed dengan post card.
- Reaksi dan jumlah reply.

Cultural Map

Elemen tampilan:

- Panel peta/list finding.
- Filter kategori.
- Author row dengan avatar.
- Badge kategori.
- Count confirmed dan reply.

Permintaan

Elemen tampilan:

- Composer request.

- Segmented control "Permintaan Terbaru" dan "Sinyal Community".
- Request card dengan author, lokasi, budget, radius, expiry.
- Demand signal card dengan kategori, jumlah request, median budget, cluster radius, status.

Detail Post

Elemen tampilan:

- Detail post.
- Media preview.
- Location map jika ada.
- Reaction bar.
- Comment thread depth terbatas.
- Report button.

Friends

Elemen tampilan:

- Tab friends, incoming, outgoing.
- Avatar dan nama user.
- Relationship action button.
- Link ke profil user.

13. Frontend API Client

File:

```
frontend/src/features/community/api/community.api.ts
```

Client memakai:

```
authenticatedFetch
NEXT_PUBLIC_API_URL
```

Jika `NEXT_PUBLIC_API_URL` kosong, client melempar error:

```
NEXT_PUBLIC_API_URL belum dikonfigurasi.
```

Fungsi client utama:

- `getCommunityFeed`
- `createCommunityPost`
- `getCommunityPost`
- `getCommunityComments`
- `createCommunityComment`
- `setCommunityReaction`
- `getCommunityCulturalMap`
- `getCommuterRequests`
- `createCommuterRequest`
- `getCommuterRequest`
- `getCommunityDemandSignals`
- `getCommunityDemandSignal`

- `getCommunityDemandSignalResponses`
- `upsertCommunityDemandSignalResponse`
- `getCommunityNotifications`
- `markCommunityNotificationRead`
- `markAllCommunityNotificationsRead`
- `createCommunityReport`
- `getCommunityReputation`
- `getCommunityAnalytics`
- `getCommunityUserProfile`
- `sendCommunityFriendRequest`
- `actOnCommunityFriendship`
- `getCommunityFriends`

14. Security Model

Security layer:

- API route memakai authenticated guard untuk user Community.
- Admin API memakai admin policy/check.
- Supabase RLS aktif pada tabel Community.
- Direct table grants dibatasi; operasi utama lewat RPC.
- Storage Community post memakai signed URL, bukan public URL.
- Avatar profile public karena bucket `avatars` memang untuk profile display.
- Service memvalidasi input dengan Zod sebelum masuk repository.
- Repository memetakan error database ke error aplikasi.

RLS/high-level policy:

- `community_posts` : authenticated read, own insert/update sesuai rule migration.
- `community_media` : authenticated read, insert untuk owner post.
- `community_comments` : authenticated read/insert.
- `community_reactions` : authenticated read/insert/delete sesuai user.
- `commuter_requests` : authenticated select, insert/update own.
- `community_notifications` : recipient select/update read state.
- `community_reports` : reporter select, create via RPC.
- `community_friendships` : participant select; mutation via RPC.
- `community_realtime_events` : scoped select berdasarkan recipient/post/signal.

15. Testing dan Verifikasi

Test backend yang berhasil dijalankan:

```
npm test
```

Hasil terakhir:

```
Test Files: 75 passed, 1 skipped
Tests: 511 passed, 1 skipped
```

File test Community yang relevan:

```

backend/tests/unit/community/community.repository.test.ts
backend/tests/unit/community/community.schema.test.ts
backend/tests/unit/community/community.service.test.ts
backend/tests/integration/community-demand-signals-route.test.ts
frontend/tests/community-page.test.ts

```

Checklist QA manual:

1. Login sebagai user biasa.
2. Buka `/community`.
3. Buat post general tanpa foto.
4. Buat finding dengan kategori dan lokasi.
5. Tambahkan emoji.
6. Upload foto via kamera/galeri.
7. Pastikan foto tampil dan signed URL valid.
8. Buka detail post.
9. Tambah comment root.
10. Tambah reply sampai depth 2.
11. Coba reaction helpful, interesting, confirmed.
12. Buka Cultural Map dan cek finding muncul.
13. Buat commuter request.
14. Buka detail commuter request.
15. Buka tab Sinyal Community.
16. Login sebagai merchant/owner yang punya merchant.
17. Buat respons UMKM untuk demand signal.
18. Cek notifikasi user target.
19. Tandai notifikasi read dan read all.
20. Laporkan post/comment.
21. Login admin dan cek report.
22. Jalankan action moderation.
23. Buka profil user lain.
24. Kirim friend request.
25. Accept/decline/cancel/unfriend sesuai role.
26. Cek avatar profile tampil di semua surface.

16. Environment dan Cara Menjalankan

Install dependency:

```
npm install
```

Menjalankan frontend:

```
npm run dev -w frontend
```

Menjalankan backend:

```
npm run dev -w backend
```

Menjalankan semua service dev:

```
npm run dev
```

Environment penting:

```
NEXT_PUBLIC_API_URL
NEXT_PUBLIC_SUPABASE_URL
NEXT_PUBLIC_SUPABASE_ANON_KEY
SUPABASE_SERVICE_ROLE_KEY
```

Backend berjalan default di port 8080 melalui script:

```
node ../scripts/dev-next.mjs backend 8080
```

Frontend berjalan default di port 3000 melalui script:

```
node ../scripts/dev-next.mjs frontend 3000
```

17. Build, Lint, dan Known Blockers

Pada audit terakhir, fitur Community tidak muncul sebagai sumber error typecheck baru. Namun gate global project belum hijau karena modul lain.

Backend Typecheck Blocker

Command:

```
npm run typecheck -w backend
```

Masih gagal pada area:

- `backend/app/api/admin/merchant-submissions/...`
- `backend/app/api/umkm/merchant-submissions/...`
- `backend/app/api/umkm/workspace/route.ts`
- `backend/app/api/umkm/advertising/campaigns/[id]/analytics/route.ts`

Pola error:

- `ZodError.errors` tidak tersedia.
- Helper response dipanggil dengan jumlah argumen salah.
- `string[]` diberikan ke parameter `string`.

Frontend Typecheck Blocker

Command:

```
npm run typecheck -w frontend
```

Masih gagal pada area:

- `frontend/src/features/merchant-submission/...`
- `frontend/src/features/umkm-workspace/...`
- `frontend/src/features/umkm-advertising/analytics/...`

Pola error:

- Client mengasumsikan envelope `{ success, data, error }` padahal type actual bukan envelope.

- `account_role` tidak ada di `UserContext` .
- Beberapa implicit any.

Frontend Build Blocker

Command:

```
npm run build -w frontend
```

Masih gagal karena modul `umkm-advertising` mengimpor hook React (`useState` , `useEffect`) ke Server Component melalui barrel export.

Contoh area:

- `frontend/src/features/umkm-advertising/hooks/use-campaigns.ts`
- `frontend/src/features/umkm-advertising/hooks/use-advertising-eligibility.ts`
- `frontend/src/features/umkm-advertising/ad-serving/hooks/use-sponsored-pin-candidates.ts`
- `frontend/app/umkm/advertising/analytics/page.tsx`

Backend Build Note

Backend build pernah terblokir karena Next mendeteksi proses build/dev lain masih berjalan dan mengunci `.next` .

Rekomendasi:

1. Matikan proses dev/build Next yang tidak dibutuhkan.
2. Bersihkan lock/cache jika aman.
3. Ulangi `npm run build -w backend` .

18. Handoff untuk Developer Selanjutnya

Prioritas sebelum release:

1. Selesaikan blocker typecheck/build/lint global di modul UMKM dan advertising.
2. Jalankan ulang:
 - `npm run typecheck -w backend`
 - `npm run typecheck -w frontend`
 - `npm run lint -w backend`
 - `npm run lint -w frontend`
 - `npm test -w backend`
 - `npm run build -w backend`
 - `npm run build -w frontend`
3. Jalankan QA manual Community dari checklist.
4. Jalankan audit responsive untuk mobile dan desktop.
5. Verifikasi RLS dengan user berbeda:
 - user A tidak bisa mengubah notifikasi user B.
 - user A tidak bisa menerima friend request milik user B.
 - merchant hanya bisa merespons memakai merchant yang dimiliki/diotorisasi.
6. Verifikasi avatar canonical setelah user mengganti foto profile.
7. Verifikasi signed URL media post tidak stale saat feed refresh.

19. File yang Paling Sering Disentuh

Jika ingin mengubah kontrak API:

```
backend/src/features/community/types/community.types.ts
backend/src/features/community/schemas/community.schema.ts
frontend/src/features/community/types/community.types.ts
frontend/src/features/community/api/community.api.ts
```

Jika ingin mengubah query database:

```
backend/src/features/community/repositories/community.repository.ts
backend/supabase/migrations/
```

Jika ingin mengubah UI:

```
frontend/src/features/community/components/
frontend/src/features/community/components/community.module.css
```

Jika ingin mengubah media upload:

```
backend/src/features/community/services/community-media.service.ts
backend/src/features/community/constants/community.constants.ts
```

Jika ingin mengubah realtime:

```
frontend/src/features/community/services/community-realtime.service.ts
backend/supabase/migrations/20260823210000_add_community_operational_layer.sql
```

20. Catatan Arsitektur

Keputusan penting:

- Business mutation Community dipusatkan pada Supabase RPC, bukan direct table mutation dari frontend.
- Frontend hanya berkomunikasi ke backend API route, bukan langsung ke table.
- Backend repository adalah satu-satunya adapter domain Community ke Supabase RPC.
- DTO frontend memakai camelCase.
- Payload API input tetap snake_case untuk beberapa field agar konsisten dengan existing backend contract.
- Avatar profile tidak digandakan di Community.
- Foto post dan foto profile dipisahkan:
 - Foto post: private-ish signed URL dari `community-media`.
 - Foto profile: canonical public avatar URL dari `profiles.avatar_url`.

21. Quick Reference

URL frontend:

```
/community
/community/[id]
/community/requests/[requestId]
/community/requests/signals/[signalId]
/community/users/[userId]
/community/friends
```

Header tab:

Beranda | Temuan Komuter | Cultural Map | Permintaan | Teman

Max content:

Post: 500 karakter
Comment: 500 karakter
Request title: 120 karakter
Request description: 500 karakter
UMKM response message: 500 karakter
Report details: 500 karakter

Media:

Input: jpeg/png/webp, max 10 MB
Output: image/webp, max 2048 px, max 10 MB
Signed URL: 10 menit

Demand:

Window: 7 hari
Cluster radius: 1000 m
Minimal request: 3
Budget bucket: Rp25.000

22. Kesimpulan

Community sudah memiliki fondasi fitur yang lengkap untuk pengalaman sosial berbasis lokasi di GETRA. Handoff ini mencakup struktur file, API, kontrak data, database, UI, realtime, media, security, testing, dan blocker release.

Sebelum release production, tim perlu menyelesaikan blocker global project di luar modul Community, lalu menjalankan ulang semua gate quality dan QA manual Community.